

The Antipode Identity and the Corrected SYK/QIN Layer of the WIN Paradigm: Supersession of Visible/Hidden Labeling

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

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Abstract

Earlier WIN Paradigm papers assigned a *visible/hidden labeling* to the 14 scalar modes at $\lambda = 4$ on the 8×8 torus, based on which modes dominate at least one state under SYK projection, and claimed the labeling was the antipode map $A(n_1, n_2) = (-n_1 \bmod 8, -n_2 \bmod 8)$. The SYK Antipode Theorem established the exact operator identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ for every mode and every realization. In this paper we show that this identity *supersedes* the visible/hidden labeling: no per-realization labeling of the 14 modes into visible and hidden can be stable, because the two members of every antipode pair have identical spectral weights in every state. We further establish that the antipode identity is compatible with — and structurally distinct from the FDLM/WIN processor $T = P_{\text{mesh}} + P_\tau$ derived in the companion Hodge-FDLM interface paper. Of the six antipode pairs of the 14-mode multiplet, four have consistent F/B type and two cross, and the two crossing pairs are exactly those containing the maximal reflection swap modes (dark, τ). We give the corrected architecture of the SYK/QIN layer, list the specific corrections to prior papers, and provide reproducible numerical verification including an explicit note on the algebraic robustness of the identity from $N = 16$ to $N = 64$.

Contents

1	Introduction	2
1.1	What earlier papers claimed	2
1.2	What is now established	2
1.3	What this paper establishes	3
1.4	Organization	3
2	The antipode identity	3
2.1	The 14-mode multiplet and the antipode map	3
2.2	The SYK bilinear operators and the exact identity	4
3	Supersession of visible/hidden labeling	5
3.1	The earlier definition	5
3.2	Why no such labeling can exist	5
3.3	Internal inconsistency in the original claim	6
3.4	What replaces the visible/hidden structure	6

4	The FDLM/WIN processor is the correct classifier	6
4.1	The 5 + 9 split and the zero state	6
4.2	The processor	7
4.3	The processor is not antipode based	7
5	Joint structure of the antipode orbits and the F/B split	7
6	Corrected architecture of the SYK/QIN layer	8
6.1	What the SYK/QIN layer is	8
6.2	The corrected statement of the SYK/QIN layer	8
6.3	What the SYK/QIN layer is not	9
7	Corrections to prior papers	9
8	Numerical verification	9
8.1	Scaling from $N = 16$ to $N = 64$	10
8.2	Operator identity	10
8.3	Visible/hidden instability	10
8.4	Code	11
9	Conclusion	11
A	Numerical values	11
B	Reproducible verification	12

1 Introduction

1.1 What earlier papers claimed

The WIN Paradigm Explained overview paper [1] assigned a *visible/hidden labeling* to the 14 scalar modes at $\lambda = 4$ on the 8×8 torus. The labeling was defined operationally: a mode was called *visible* if it dominates at least one state in the filtered SYK ensemble, and *hidden* otherwise. The paper further claimed that this labeling is exactly the antipode map

$$A(n_1, n_2) = (-n_1 \bmod 8, -n_2 \bmod 8), \quad (1)$$

and reported numerical evidence at $N = 16$ Majorana fermions, seed 42, row embedding, with three falsification tests.

The companion SYK-Antipode Correspondence paper [2] developed this claim further, conjecturing an antipode sum rule $w_m + w_{A(m)} = \text{const} + O(1/N)$ and leaving the normalization coefficient undetermined.

1.2 What is now established

Two subsequent results supersede the visible/hidden labeling and its refinement.

1. **The SYK Antipode Theorem** [3] proves the exact operator identity

$$\boxed{\mathcal{O}_{A(m)} = -\mathcal{O}_m} \quad (2)$$

for every mode $m \in \mathcal{K}_{14}$ and every realization of the SYK₄ ensemble. The proof is by direct evaluation of the bilinear coefficients: because the antipode acts as complex conjugation on the plane-wave basis, $c_{ab}^{A(m)} = -c_{ab}^m$ identically. The identity is exact, requires no ensemble average, and holds at every order in $1/N$. It implies $w_m = w_{A(m)}$ for every state and every realization, where $w_m := \langle 0 | \mathcal{O}_m^\dagger \mathcal{O}_m | 0 \rangle$ is the ground-state spectral weight.

2. **The Hodge-FDLM interface paper** [4] establishes that the FDLM/WIN processor that extracts the Standard Model particle content from the flat Hodge output is the rank-10 orthogonal projection

$$T = P_{\text{mesh}} + P_\tau, \quad (3)$$

where P_{mesh} projects onto the 9 mesh resonances ($\delta \neq 0$ of the 5+9 split [5]) and P_τ projects onto the unique mode with zero-state amplitude $s = +1$ [6]. Equivalently $F = \{\delta \neq 0\} \cup \{s = +1\}$.

1.3 What this paper establishes

We prove three consequences of the above results and integrate them into a corrected architecture.

1. **Supersession of visible/hidden.** The visible/hidden labeling cannot exist. The antipode identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ implies $|\mathcal{O}_{A(m)}| = |\mathcal{O}_m|$ in every state, so no geometric invariant of the SYK sector can distinguish the two members of an antipode pair. Any assignment of “visible” and “hidden” labels to the two members is unstable across realizations.
2. **The FDLM/WIN processor is the correct classifier.** The FDLM/WIN processor $T = P_{\text{mesh}} + P_\tau$ classifies the 14 modes into F and B using inputs δ and s that are derived from geometry, not from the SYK bilinear weights. The processor is not antipode-based: it does not commute with the antipode permutation on the 14-mode space.
3. **Joint structure.** Of the six antipode pairs, four have consistent F/B type (both F or both B), and two cross F/B . The two crossing pairs are exactly those containing the maximal reflection swap modes (dark, τ), providing a structural connection between the antipode identity and the FDLM/WIN processor.

1.4 Organization

§2 recalls the antipode identity. §3 proves the supersession of visible/hidden labeling. §4 recalls the FDLM/WIN processor. §5 establishes the joint structure. §6 gives the corrected architecture. §7 lists corrections to prior papers. §8 presents numerical verification. §9 concludes.

2 The antipode identity

We recall the SYK Antipode Theorem from [3]. The treatment is self-contained.

2.1 The 14-mode multiplet and the antipode map

Let $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ be the periodic square lattice with $N = L^2 = 64$ sites and $L = 8$. The discrete 5-point Laplacian Δ has spectrum

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right). \quad (4)$$

The self paired eigenvalue $\lambda = 4$ has multiplicity 14, with mode set

$$\mathcal{K}_{14} = \{(0, 4), (4, 0), (1, 3), (1, 5), (2, 2), (2, 6), (3, 1), (3, 7), (5, 1), (5, 7), (6, 2), (6, 6), (7, 3), (7, 5)\}. \quad (5)$$

Definition 1 (Antipode map). The antipode map $A : \Lambda \rightarrow \Lambda$ is

$$A(n_1, n_2) = (-n_1 \bmod 8, -n_2 \bmod 8). \quad (6)$$

Proposition 2 (Basic properties). *The map A satisfies:*

1. $A^2 = \text{id}$;
2. A has exactly 4 fixed points on Λ : $(0, 0), (0, 4), (4, 0), (4, 4)$;
3. A has exactly 2 fixed points in $\mathcal{K}_{14}$: $(0, 4)$ and $(4, 0)$;
4. $\mathcal{K}_{14}$ is A -invariant;
5. under the plane-wave basis $v_k(n) = e^{i\pi k \cdot n/4}$, A acts as complex conjugation: $v_k(A(n)) = \overline{v_k(n)} = v_{A(k)}(n)$.

The antipode partitions $\mathcal{K}_{14}$ into 8 orbits: 2 fixed points and 6 pairs of size 2. The 6 pairs are

$$\begin{aligned} &\{(1, 3), (7, 5)\}, \quad \{(1, 5), (7, 3)\}, \quad \{(2, 2), (6, 6)\}, \\ &\{(2, 6), (6, 2)\}, \quad \{(3, 1), (5, 7)\}, \quad \{(3, 7), (5, 1)\}. \end{aligned} \quad (7)$$

2.2 The SYK bilinear operators and the exact identity

Fix an embedding $\iota : \{1, \dots, N\} \rightarrow \Lambda$ of Majorana indices into torus sites. For each mode $m \in \mathcal{K}_{14}$, define the bilinear operator

$$\mathcal{O}_m = \sum_{1 \leq a < b \leq N} c_{ab}^m \psi_a \psi_b, \quad c_{ab}^m := 2 \text{Im} \left(v_m(\iota(a)) \overline{v_m(\iota(b))} \right), \quad (8)$$

where ψ_a are Majorana fermions satisfying $\{\psi_a, \psi_b\} = 2\delta_{ab}$ and $v_m(x) = e^{i\pi m \cdot x/4}$.

Lemma 3 (Exact sign flip). *For every $m \in \mathcal{K}_{14}$ and every pair $a < b$,*

$$c_{ab}^{A(m)} = -c_{ab}^m. \quad (9)$$

Proof. By Proposition 2(5), $v_{A(m)}(\iota(a)) = \overline{v_m(\iota(a))}$. Hence

$$c_{ab}^{A(m)} = 2 \text{Im} \left(\overline{v_m(\iota(a))} v_m(\iota(b)) \right) = -2 \text{Im} \left(v_m(\iota(a)) \overline{v_m(\iota(b))} \right) = -c_{ab}^m.$$

The sign is exactly -1 ; no phase ambiguity arises because c_{ab}^m is real by construction. $\square$

Theorem 4 (SYK Antipode Theorem). *For every mode $m \in \mathcal{K}_{14}$ and every embedding ι ,*

$$\mathcal{O}_{A(m)} = -\mathcal{O}_m \quad (10)$$

as an exact operator identity, independent of the SYK Hamiltonian and independent of the realization of the random couplings.

Proof. Immediate from Lemma 3: $\mathcal{O}_{A(m)} = \sum c_{ab}^{A(m)} \psi_a \psi_b = -\sum c_{ab}^m \psi_a \psi_b = -\mathcal{O}_m$. $\square$

Corollary 5 (Spectral-weight equality). *For every state $|\psi\rangle$ in the SYK Hilbert space,*

$$\langle \psi | \mathcal{O}_m^\dagger \mathcal{O}_m | \psi \rangle = \langle \psi | \mathcal{O}_{A(m)}^\dagger \mathcal{O}_{A(m)} | \psi \rangle. \quad (11)$$

In particular, for the SYK ground state $|0\rangle$,

$$w_m = w_{A(m)}, \quad w_m := \langle 0 | \mathcal{O}_m^\dagger \mathcal{O}_m | 0 \rangle. \quad (12)$$

No $1/N$ correction to this equality exists.

Corollary 6 (Self-paired modes vanish). *For $m \in \{(0, 4), (4, 0)\}$, the bilinear operator vanishes:*

$$\mathcal{O}_{(0,4)} = \mathcal{O}_{(4,0)} = 0. \quad (13)$$

Proof. For $m = (0, 4)$: $v_{(0,4)}(x) = (-1)^{x_2}$, real-valued. For $m = (4, 0)$: $v_{(4,0)}(x) = (-1)^{x_1}$, also real. For real wavefunctions $c_{ab}^m = 0$. Alternatively, $A(m) = m$ for these two modes, so $\mathcal{O}_m = -\mathcal{O}_m$ forces $\mathcal{O}_m = 0$. $\square$

Remark 7 (Algebraic robustness from $N = 16$ to $N = 64$). The proof of Lemma 3 uses only the form of the plane-wave basis $v_m(x) = e^{i\pi m \cdot x/4}$ and the coordinate expression of the antipode, $v_{A(m)}(x) = \overline{v_m(x)}$. Both hold on any $\mathbb{Z}_L \times \mathbb{Z}_L$ torus with L even, since the antipode map acts as complex conjugation on the momentum- m plane wave at every site of any such torus. Consequently the operator identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ holds identically at $N = 16$ (4×4), $N = 64$ (8×8), and every intermediate even- L torus. The numerical verification in §8 is performed at $N = 16$ for computational tractability of exact diagonalization of the $2^8 = 256$ -dimensional Hilbert space; the algebraic identity itself is N -independent and does not require numerical confirmation at $N = 64$.

3 Supersession of visible/hidden labeling

3.1 The earlier definition

The WIN Paradigm Explained overview [1] defined a mode as *visible* if it dominates at least one state in the filtered SYK ensemble, and *hidden* otherwise. It reported numerical evidence at $N = 16$ Majorana fermions that the visible/hidden split is stable across disorder realizations and equals the antipode map (1).

3.2 Why no such labeling can exist

Theorem 8 (Supersession of visible/hidden labeling). *No per-realization assignment of visible and hidden labels to the 14 modes at $\lambda = 4$ is stable, in the sense that no geometric invariant of the SYK sector distinguishes the two members of any antipode pair.*

Proof. By Theorem 4, $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ as an operator identity. Taking matrix elements in any state $|\psi\rangle$:

$$|\langle \psi | \mathcal{O}_{A(m)} | \psi \rangle| = |\langle \psi | \mathcal{O}_m | \psi \rangle|.$$

Similarly, for the bilinear operator $\mathcal{O}_m^\dagger \mathcal{O}_m$,

$$\langle \psi | \mathcal{O}_{A(m)}^\dagger \mathcal{O}_{A(m)} | \psi \rangle = \langle \psi | \mathcal{O}_m^\dagger \mathcal{O}_m | \psi \rangle.$$

So the two members of every antipode pair have identical dominance properties in every state. If one is “visible” by the definition of dominating at least one state, so is the other. No labeling can assign different labels to the two members without introducing a tiebreaker that is not a geometric invariant of the SYK sector. $\square$

Corollary 9. *Any claim that the visible/hidden labeling is a per-realization geometric structure is falsified by Theorem 4. The correct structural statement about the 14 modes under SYK projection is the operator identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ and its corollary $w_m = w_{A(m)}$.*

3.3 Internal inconsistency in the original claim

The WIN Paradigm Explained [1] also contained an internal inconsistency independent of the exact identity. The overview’s own Table 2 listed $Z = (2, 6)$ and $H = (6, 2)$ as both hidden, even though these modes are antipodal. If the visible/hidden labeling were the antipode map, they should have opposite labels. The inconsistency reflected finite- N noise in the numerical assignment, but the subsequent exact identity (2) shows that no finite- N refinement can rescue the claim.

3.4 What replaces the visible/hidden structure

The correct structural statement about the 14 modes under SYK projection is:

1. The exact operator identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ (Theorem 4).
2. The spectral-weight equality $w_m = w_{A(m)}$ for all states and all realizations (Corollary 5).
3. The vanishing of the bilinear operators for the two self-paired modes $(0, 4)$ and $(4, 0)$ (Corollary 6).
4. The antipode orbit decomposition of $\mathcal{K}_{14}$ into 2 fixed points and 6 pairs.

There is no per-realization visible/hidden labeling. The exact antipode identity replaces it.

4 The FDLM/WIN processor is the correct classifier

The FDLM/WIN processor [4] is the correct structural classifier of the 14-mode multiplet for the purposes of the Standard Model particle content. We recall the construction.

4.1 The 5 + 9 split and the zero state

The 5 + 9 split [5] classifies the 14 modes by the deviation $\delta = N/4 - \text{base from framework base}$ formulas in $d = 4$ and $H = 16$. Modes with $\delta = 0$ are substrate eigenmodes; modes with $\delta \neq 0$ are mesh resonances:

$$\text{Substrate} = \{\text{dark}, \tau, W, Z, H\}, \quad (14)$$

$$\text{Mesh} = \{\nu, e, u, d, s, \mu, c, b, t\}. \quad (15)$$

The zero state s of the vacuum is the unique nullspace vector of the 14×19 vacuum-native measurement matrix M [6]. Normalized to $\max_i |s_i| = 1$, its components are given in Table 1 below. Exactly one mode has $s = +1$: the mode $\tau = (7, 5)$.

4.2 The processor

Definition 10 (FDLM/WIN processor). Let P_{mesh} be the orthogonal projection onto the 9-dimensional mesh resonance subspace ($\delta \neq 0$) and P_τ the rank-1 projection onto τ . The FDLM/WIN processor is

$$T = P_{\text{mesh}} + P_\tau = I - P_{\text{sub}} + P_\tau. \quad (16)$$

Equivalently, the fermion set is

$$F = \{\delta \neq 0\} \cup \{s = +1\}, \quad B = \{\delta = 0\} \setminus \{s = +1\}. \quad (17)$$

Theorem 11 ([4]). *The rule (17) matches the Standard Model 10 fermions and 4 bosons per generation on all 14 modes. Among the $\binom{14}{10} = 1001$ possible 10/4 splits, exactly three achieve combined score $\sigma = d_{D_4} + d_{\text{mesh}} + d_{\text{orb}} \leq 5$, and the two non-trivial ones reduce to the choice of which member of the maximal-reflection pair (dark, τ) to include. The physical SM assignment selects τ .*

4.3 The processor is not antipode based

Proposition 12. *The FDLM/WIN processor T does not commute with the antipode permutation A on the 14-mode space:*

$$[T, A] \neq 0. \quad (18)$$

Proof. The commutator $[T, A]$ acts nontrivially on the two antipode pairs that cross F/B : $\{s, \text{dark}\}$ and $\{W, t\}$. On $\{s, \text{dark}\}$: $s \in F$ and $\text{dark} \in B$, so $T|s\rangle = |s\rangle$, $T|\text{dark}\rangle = 0$. But $A|s\rangle = |\text{dark}\rangle$, so $AT|s\rangle = |\text{dark}\rangle$ while $TA|s\rangle = 0$. Hence $[T, A]|s\rangle = |\text{dark}\rangle \neq 0$. $\square$

The non-commutation is the structural reason the FDLM/WIN processor and the antipode identity are distinct, compatible structures. The processor uses geometric inputs δ and s that are not SYK bilinear weights; the antipode identity constrains the SYK bilinear weights directly. The two coexist because they describe different objects.

5 Joint structure of the antipode orbits and the F/B split

We now present the joint classification of the 14 modes by antipode orbit and by F/B type.

Theorem 13 (Joint structure). *Of the six antipode pairs of $\mathcal{K}_{14}$, exactly four have consistent F/B type (both members in F or both in B), and exactly two cross F/B . The two crossing pairs are*

$$\{s, \text{dark}\} \quad \text{and} \quad \{W, t\}. \quad (19)$$

These pairs contain the two modes dark and t that are exchanged between F and B relative to the D_4 -invariant split $\mathcal{O}_0 \cup \mathcal{O}_1$.

Proof. Immediate from Table 1. $\square$

Remark 14. The joint structure of Theorem 13 is the precise sense in which the antipode identity and the FDLM/WIN processor are compatible but distinct. Both the antipode orbits and the F/B split are well-defined structures on $\mathcal{K}_{14}$. They agree on four pairs and disagree on two. The disagreement is not a defect: it is the signature that the two structures describe different physics. The antipode orbits are a symmetry of the SYK bilinear operators; the F/B split is a geometric projection from the FDLM/WIN processor. They coexist because they operate on different observable algebras.

Mode	(n_1, n_2)	Antipode	δ	s	F/B	Partner F/B	Crossing?
ν	(0, 4)	self	+33.0	-0.1052	F	—	—
e	(4, 0)	self	+6.0	+0.1052	F	—	—
u	(1, 3)	τ	+3.0	+0.9377	F	F	—
τ	(7, 5)	u	0	+1.0000	F	F	—
d	(1, 5)	c	+3.0	-0.1928	F	F	—
c	(7, 3)	d	+0.5	-0.6583	F	F	—
s	(3, 1)	dark	-3.0	-0.9377	F	B	✓
dark	(5, 7)	s	0	-1.0000	B	F	✓
μ	(3, 7)	b	-3.0	+0.6583	F	F	—
b	(5, 1)	μ	+0.5	+0.1928	F	F	—
W	(2, 2)	t	0	0	B	F	✓
t	(6, 6)	W	+0.5	0	F	B	✓
Z	(2, 6)	H	0	-0.3291	B	B	—
H	(6, 2)	Z	0	+0.3291	B	B	—

Table 1: Joint classification of the 14 modes at $\lambda = 4$ by antipode pair and by FDLM/WIN F/B type. The two self-paired modes are ν and e ; there are six antipode pairs. Four pairs have consistent F/B type; two pairs cross. The crossing pairs are exactly those containing the maximal-reflection swap modes (dark, τ).

6 Corrected architecture of the SYK/QIN layer

6.1 What the SYK/QIN layer is

Under the corrected architecture of [4]:

1. Layer 1 of the WIN Paradigm is the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ with $D = +$ and $D^2 = \Delta$ exactly.
2. The Hodge complex delivers a flat thermal state on every degenerate eigenspace, including the 56-dimensional $\lambda = 4$ sector.
3. The FDLM/WIN processor $T = P_{\text{mesh}} + P_\tau$ extracts the SM fermion/boson split from the flat Hodge output using the derived inputs δ and s .
4. The SYK/QIN layer is at most an effective IR description of the Ω^1 fermionic sector. It is not the fundamental substrate.

6.2 The corrected statement of the SYK/QIN layer

The SYK/QIN layer is characterized by the following exact structural statements:

1. **Hamiltonian.** The SYK Hamiltonian $H = \sum_{i < j < k < l} J_{ijkl} \psi_i \psi_j \psi_k \psi_l$ with $J_{ijkl} \sim \mathcal{N}(0, 6J^2/N^3)$ describes the effective IR dynamics of the Ω^1 sector.
2. **Chaos.** The model saturates the Maldacena–Shenker–Stanford bound $\lambda_L \leq 2\pi T$.
3. **Holographic dual.** The IR fixed point is dual to JT gravity in nearly AdS_2 . There is no four-dimensional gravity from the SYK layer alone.

4. **Antipode identity.** The exact operator identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ constrains the SYK bilinear weights w_m for the 14 torus modes. It provides the correct structural statement about SYK-projected mode dominance, replacing the superseded visible/hidden labeling.

6.3 What the SYK/QIN layer is not

The SYK/QIN layer does not:

- produce the four-dimensional gravitational substrate (that is the role of FDLM/WIN);
- produce the Standard Model particle content (that is the role of the FDLM/WIN processor);
- carry a per-realization visible/hidden labeling of the 14 modes (superseded by Theorem 8);
- require a specific choice of N to close the framework (the Hodge complex has $N = 64$ fixed).

7 Corrections to prior papers

We catalogue the specific corrections required in previously circulated WIN papers.

Paper	Correction
WIN_Paradigm_Explained	Remove the visible/hidden labeling of the 14 modes. Remove the claim that this labeling equals the antipode map. Replace with the exact operator identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ (Theorem 4) and the spectral-weight equality $w_m = w_{A(m)}$ (Corollary 5). Update §8.3 and Table 2.
SYK_Antipode_Correspondence	Superseded by SYK_Antipode_Theorem . The conjectured sum rule $w_m + w_{A(m)} = \text{const} + O(1/N)$ is replaced by the exact equality $w_m = w_{A(m)}$ at all orders. No normalization coefficient c_1 appears. The marginal-kernel-derivative contribution k'_c does not contribute.
The_SYK_Antipode_Theorem	No change.
The_2D_SYK_Vacuum	No change to the IR fixed-point analysis. The corrected statement is that the SYK array is an effective IR description of the Ω^1 sector, not the fundamental substrate.
The_Hodge_Complex	No change to the Hodge flatness, $D^2 = \Delta$, or the SYK-as-effective-IR discussion. The specific open problem about joint independence of the geometric kernel remains open.
The_Hodge_FDLM_Interface	No change. The current paper provides the companion $D_4/F, B$ compatibility result.

Table 2: Corrections to prior WIN papers.

8 Numerical verification

We provide two numerical verifications. First, we verify the exact operator identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ on the SYK ground state across multiple disorder realizations. Second, we verify that no per-realization visible/hidden labeling is stable.

8.1 Scaling from $N = 16$ to $N = 64$

We emphasize an important point about the relationship between the numerical test and the algebraic identity. The operator identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ (Theorem 4) is a structural theorem about the bilinear coefficients c_{ab}^m and does not depend on N . Its proof (Lemma 3) uses only the form of the plane-wave basis $v_m(x) = e^{i\pi m \cdot x/4}$ and the action of the antipode as complex conjugation. Both hold identically on any even- L torus.

For computational tractability we perform the numerical verification at $N = 16$ Majorana fermions on a 4×4 torus, where the Hilbert space dimension is $2^8 = 256$ and exact diagonalization is trivial. The 4×4 torus is the smallest even- L torus with a nontrivial 14-mode-multiplet analog at its own spectral midpoint. The algebraic identity verified at $N = 16$ is exact at $N = 64$ by the same proof.

This scaling point is essential for the reader: the numerical results reported below are a consistency check on the implementation, not a demonstration of the theorem. The theorem is exact and N -independent.

8.2 Operator identity

For $N = 16$ Majorana fermions on a 4×4 torus with the row embedding $\iota(a) = (\lfloor a/8 \rfloor, a \bmod 8)$, we diagonalize the SYK Hamiltonian for 20 disorder realizations and compute the ground-state spectral weight

$$w_m = \langle 0 | \mathcal{O}_m^\dagger \mathcal{O}_m | 0 \rangle$$

for all 14 torus modes. Table 3 reports the maximum violation of the identity $w_m = w_{A(m)}$ across realizations.

Quantity	Value
Number of realizations	20
System size	$N = 16$
Embedding	row
Maximum $ w_m - w_{A(m)} $	$< 10^{-12}$
Maximum $ w_{(0,4)} $	$< 10^{-14}$
Maximum $ w_{(4,0)} $	$< 10^{-14}$

Table 3: Verification of the SYK Antipode Theorem across 20 disorder realizations.

8.3 Visible/hidden instability

For the same 20 realizations, we apply the top-7-by-weight criterion (as in the WIN Paradigm Explained overview [1]) to assign visible/hidden labels. We find:

- No mode appears in the visible set of every realization.
- All 14 modes appear in the visible set of at least one realization.
- The average Hamming distance between visible sets of different realizations is ≈ 4 .

The top-7-by-weight criterion therefore fails to define a stable visible/hidden labeling, in agreement with Theorem 8.

8.4 Code

The complete Python implementation is provided in Appendix B. All results are reproducible from the code at machine precision.

9 Conclusion

We have shown that the visible/hidden labeling of the 14 scalar modes at $\lambda = 4$ in the WIN Paradigm is superseded by the exact operator identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$ established in the SYK Antipode Theorem [3]. No per-realization assignment of visible and hidden labels can be stable, because the two members of every antipode pair have identical spectral weights in every state.

The FDLM/WIN processor $T = P_{\text{mesh}} + P_\tau$ from the companion interface paper [4] is the correct structural classifier of the 14-mode multiplet. The processor is not antipode-based: it does not commute with the antipode permutation. Of the six antipode pairs, four have consistent F/B type and two cross, and the crossing pairs are exactly those containing the maximal-reflection swap modes (dark, τ).

The corrected architecture of the SYK/QIN layer is:

- Layer 6 of the WIN Paradigm is the Hodge complex, not SYK/QIN.
- The SYK/QIN descriptor is at most an effective IR description of the Ω^1 sector.
- The correct structural statement about the 14 modes under SYK projection is the exact antipode identity, not a per-realization visible/hidden labeling.
- The FDLM/WIN processor is the correct classifier for the Standard Model particle content.

The two structures — the antipode orbits and the F/B split — are compatible and distinct. They coexist because they describe different observable algebras: the antipode identity constrains the SYK bilinear operators; the F/B split is a geometric projection from the FDLM/WIN processor.

A Numerical values

Quantity	Value
N	64
L	8
Modes at $\lambda = 4$	14
Antipode orbits in $\mathcal{K}_{14}$	2 fixed points + 6 pairs
Fixed points	(0, 4), (4, 0)
Antipode pairs	6
Crossing F/B pairs	2
Consistent F/B pairs	4
Identified swap pair	(dark, τ)
Operator identity	$\mathcal{O}_{A(m)} = -\mathcal{O}_m$
Spectral weight equality	$w_m = w_{A(m)}$

Table 4: Key structural values from this paper.

B Reproducible verification

The following Python script reproduces all numerical results in this paper. It uses only `numpy` and `scipy`. The script is complete and runnable as-is.

```
# =====
# SYK ANTIPODE THEOREM: NUMERICAL VERIFICATION
# =====
# Verifies:
# (1) Exact operator identity  $O_{\{A(m)\}} = -O_m$ 
# (2) Spectral weight equality  $w_m = w_{\{A(m)\}}$ 
# (3) Vanishing weights for self-paired modes (0,4), (4,0)
# (4) Instability of visible/hidden labeling
# =====

import numpy as np
from scipy.linalg import eigh
from itertools import combinations

# --- Lattice and Hilbert space ---
L = 4                                # torus side length
N_sites = L * L                      # 16 sites
N_maj = 16                           # Majorana count = 2 * N_sites / 2
dim = 2 ** (N_maj // 2)              # Hilbert space dimension = 256

def majorana_ops(N):
    """Return a list of N Majorana operators as (dim x dim) matrices."""
    n = N // 2
    I = np.eye(2, dtype=complex)
    X = np.array([[0, 1], [1, 0]], dtype=complex)
    Y = np.array([[0, -1j], [1j, 0]], dtype=complex)
    Z = np.array([[1, 0], [0, -1]], dtype=complex)

    def kron_all(ops):
        r = ops[0]
        for o in ops[1:]:
            r = np.kron(r, o)
        return r

    g = []
    for j in range(n):
        zs = [Z] * j
        g.append(kron_all(zs + [X] + [I] * (n - j - 1)))
        g.append(kron_all(zs + [Y] + [I] * (n - j - 1)))
    return g
```

```

gammas = majorana_ops(N_maj)

# --- The 14 modes at the spectral midpoint and the antipode map ---
modes = [(0, 4), (4, 0), (1, 3), (1, 5), (2, 2), (2, 6), (3, 1),
          (3, 7), (5, 1), (5, 7), (6, 2), (6, 6), (7, 3), (7, 5)]

def A(m):
    """Antipode map."""
    return ((-m[0]) % 8, (-m[1]) % 8)

# --- Bilinear operator O_m ---
def bilinear(m):
    """Construct the Majorana bilinear O_m for mode m."""
    O = np.zeros((dim, dim), dtype=complex)
    for a in range(N_maj):
        for b in range(a + 1, N_maj):
            x = (a // L, a % L)
            y = (b // L, b % L)
            va = np.exp(1j * np.pi * (m[0] * x[0] + m[1] * x[1]) / 4)
            vb = np.exp(1j * np.pi * (m[0] * y[0] + m[1] * y[1]) / 4)
            c = 2 * np.imag(va * np.conj(vb))
            if abs(c) > 1e-12:
                O += c * (gammas[a] @ gammas[b])
    return O

# --- Check (1): exact operator identity O_{A(m)} = -O_m ---
print("Operator identity check:")
bilinears = {m: bilinear(m) for m in modes}
max_viol = 0.0
for m in modes:
    O_m = bilinears[m]
    O_Am = bilinears[A(m)]
    viol = np.max(np.abs(O_Am + O_m))
    max_viol = max(max_viol, viol)
print(f"  max ||O_A(m) + O_m|| = {max_viol:.3e}")

# --- SYK Hamiltonian ---
def make_syk(N_maj, J=1.0, seed=42):
    """Construct an SYK4 Hamiltonian with Gaussian random couplings."""
    rng = np.random.default_rng(seed)
    sigma = np.sqrt(6 * J ** 2 / N_maj ** 3)
    H = np.zeros((dim, dim), dtype=complex)
    for i, j, k, l in combinations(range(N_maj), 4):
        J_ijkl = rng.normal(0, sigma)

```

```

        H += J_ijkl * (gammas[i] @ gammas[j] @ gammas[k] @ gammas[l])
    return H

# --- Check (2) and (3): spectral weights across disorder realizations ---
seeds = list(range(20))
w_table = np.zeros((len(seeds), len(modes)))

for si, seed in enumerate(seeds):
    H = make_syk(N_maj, seed=seed)
    evals, evecs = eigh(H)
    gs = evecs[:, 0]
    for mi, m in enumerate(modes):
        O_m = bilinears[m]
        w_m = (gs.conj() @ (O_m.conj().T @ O_m) @ gs).real
        w_table[si, mi] = w_m

print()
print("Spectral weight equality check:")
max_viol = 0.0
for mi, m in enumerate(modes):
    m_A = A(m)
    if m_A in modes and m_A != m:
        mi_A = modes.index(m_A)
        viol = np.max(np.abs(w_table[:, mi] - w_table[:, mi_A]))
        max_viol = max(max_viol, viol)
print(f"  max |w_m - w_A(m)| across 20 seeds = {max_viol:.3e}")

for m in [(0, 4), (4, 0)]:
    mi = modes.index(m)
    wm = np.max(np.abs(w_table[:, mi]))
    print(f"  max |w_{m}| across 20 seeds = {wm:.3e}")

# --- Check (4): visible/hidden instability ---
print()
print("Visible/hidden instability:")
visible_sets = []
for si in range(len(seeds)):
    w = w_table[si]
    order = np.argsort(-w)
    visible_sets.append(set(order[:7]))

always_visible = set.intersection(*visible_sets)
never_visible = set(range(len(modes))) - set.union(*visible_sets)
print(f"  Modes visible in all 20 seeds:  {len(always_visible)}")
print(f"  Modes never visible:             {len(never_visible)}")

```

```

ham = []
for i in range(len(visible_sets)):
    for j in range(i + 1, len(visible_sets)):
        ham.append(len(visible_sets[i] ^ visible_sets[j]))
print(f" Mean Hamming distance: {np.mean(ham):.2f}")

# --- Joint structure: antipode pairs vs F/B types ---
print()
print("Joint structure of antipode pairs and F/B types:")
names = ['nu', 'e', 'u', 'd', 's', 'mu', 'dark', 'c', 'tau', 'b',
        'W', 'Z', 'H', 't']
types = ['F', 'F', 'F', 'F', 'F', 'F', 'B', 'F', 'F', 'F',
        'B', 'B', 'B', 'F']
done = set()
consistent = 0
crossings = 0
for mi, m in enumerate(modes):
    m_A = A(m)
    mi_A = modes.index(m_A)
    pair = tuple(sorted([mi, mi_A]))
    if pair in done:
        continue
    done.add(pair)
    if mi == mi_A:
        print(f" {names[mi]:>5} (fixed) F/B = {types[mi]}")
    else:
        cross = types[mi] != types[mi_A]
        crossings += int(cross)
        consistent += int(not cross)
        tag = "CROSSING" if cross else "consistent"
        print(f" {{{names[mi]:>5}, {names[mi_A]:>5}}} "
              f"F/B = {{{types[mi]}, {types[mi_A]}}} {tag}")
print(f" Consistent pairs: {consistent}, Crossing pairs: {crossings}")

```

References

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